

Project Report

Heritage Treasures: An In-Depth Analysis of UNESCO

World Heritage Sites in Tableau

Team ID : LTVIP2026TMIDS55764

Team Size : 4

Team Leader : Akki Baby Avanthi Gayathri

Team members : Chavali Madhava Sai

Revanth Venkata Sai Anigandla

Praveen Raj

1. INTRODUCTION

1.1 Project Overview

The UNESCO World Heritage Sites Dashboard is a data visualization project that analyses global heritage sites using interactive charts and maps. The project presents insights about site distribution by region, country, category (Cultural, Natural, Mixed), year of inscription, area (hectares), and danger status.

The dashboard is developed using Tableau for visualization and integrated into a Flask-based web application, providing an interactive and user-friendly platform for exploring UNESCO heritage data.

1.2 Purpose

The purpose of this project is to:

- Analyse and visualize UNESCO World Heritage Sites data.
- Identify regional and country-wise distribution patterns.
- Understand trends in site inscriptions over the years.
- Examine the status of sites (In Danger vs Not in Danger).
- Provide an interactive tool for academic and analytical use.

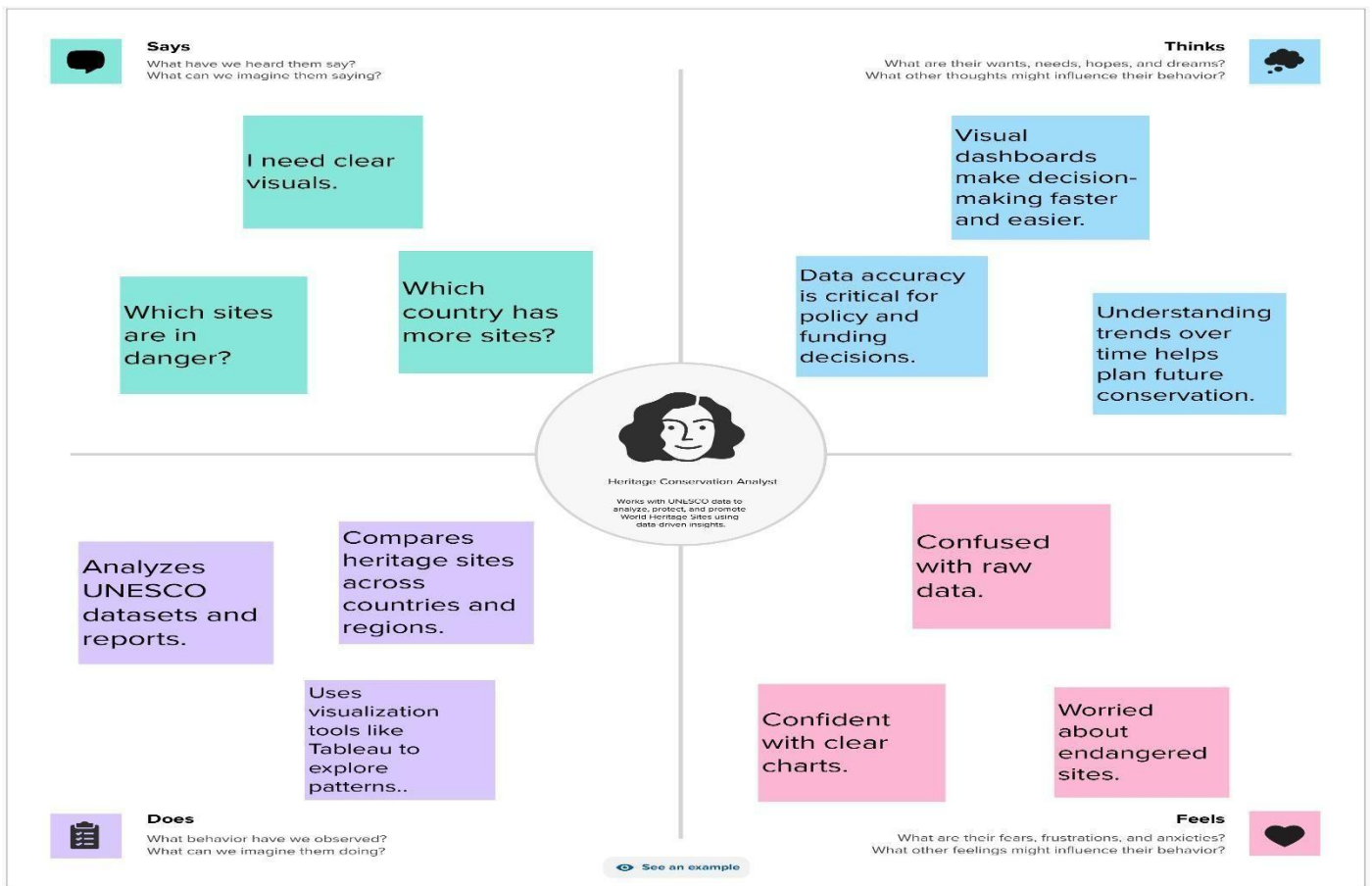
2. IDEATION PHASE

2.1 Problem Statement

	I am	I'm trying to	But	Because	Which makes me feel
	I am a heritage conservation researcher and policymaker.	I'm trying to understand how UNESCO World Heritage Sites are distributed across different countries.	But the raw data is complex and spread across large datasets without clear visual representation.	Because it is difficult to quickly identify which countries have the highest concentration of heritage sites.	Which makes me feel overwhelmed and limits my ability to prioritize preservation and cultural promotion efforts effectively.
Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	I am a heritage conservation researcher and policymaker.	I'm trying to understand how UNESCO World Heritage Sites are distributed across different countries.	But the raw data is complex and spread across large datasets without clear visual representation .	Because it is difficult to quickly identify which countries have the highest concentration of heritage sites.	Which makes me feel overwhelmed and limits my ability to prioritize preservation and cultural promotion efforts effectively.
PS-2	I am a UNESCO stakeholder and heritage management authority.	I'm trying to identify World Heritage Sites that are at risk and analyse regional trends in site inscriptions over time.	But the information is not presented in an intuitive or comparative visual format.	Because assessing endangered sites and longterm regional conservation trends requires manual analysis of raw data.	Which makes me feel uncertain about where to focus immediate conservation actions and future heritage planning strategies.

2.2 Empathy Map Canvas

Empathy Map for the project title: Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau



2.3 Brainstorming

Brainstorm & idea prioritization

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

🕒 10 minutes to prepare
🕒 1 hour to collaborate
👥 3-5 people recommended

Before you collaborate

A little bit of preparation goes a long way with this session. Here's what you need to do to get going.

🕒 50 minutes

- Team gathering**
Define who should participate in the session and send an invite. Select a facilitator and moderator to get ideas started.
- Set the goal**
Think about the problem you'll be focusing on solving in the brainstorming session.
- Know how to use the facilitation Superpowers**
Use the Facilitation Superpowers to run a happy and productive session.

[Open article](#)

1 Define your problem statement

What problem are you trying to solve? Frame your problem as a How Might We statement. This will be the focus of your brainstorm.

🕒 5 minutes

How Might We

Problem: UNESCO World Heritage sites are disappearing at an alarming rate. How might we protect and promote these sites using data-driven insights?

2 Key rules of brainstorming

To run an smooth and productive session:

- Stay in scope.
- Encourage wild ideas.
- Defend judgement.
- Focus on others.
- Go for volume.
- If possible, be visual.

Need some inspiration?

Use a Facilitation Superpower to generate your ideas.

[Open example](#)

app.mural.co/t/project04371/m/project04371/1770704602624/10404d0ba90a4298a0d03348bbbc3db976651c79

Brainstorm and idea prioritization

Facilitate Share ? S

Brainstorm
Write down any ideas that come to mind that address your problem statement.
10 minutes

Group ideas
Take turns sharing your ideas while clustering similar or related notes as you go. Once all sticky notes have been grouped, give each cluster a sentence-like label. If a cluster is bigger than six sticky notes, try and see if you can break it up into smaller sub-groups.
20 minutes

Prioritize
Your team should all be on the same page about what's important moving forward. Place your ideas on the grid to determine which ideas are important and which are feasible.
20 minutes

Avanthi
Identify top countries with highest number of UNESCO sites
Analyze heritage sites listed on category (Cultural, Natural, Mixed)
1. Create Tableau dashboard showing world map of heritage sites

Madhav
Compare heritage sites across continents
Find which year had the highest number of site additions
Analyze endangered heritage sites

Revanth
Study tourism impact on heritage sites
Identify most famous heritage sites in India
Analyze heritage sites based on historical period

Praveen
Create interactive map in Tableau (Country, Type, Year)
Design charts like bar chart, pie chart, map
Find trends and patterns in heritage site distribution

Heritage Site Distribution

Category Analysis

Time-based Analysis

Visualization and Dashboard

Special Analysis

Heritage Site Distribution

Category Analysis

Time-based Analysis

Importance

If each of these ideas could get done without any difficulty or cost, which would have the most positive impact?

12%

app.mural.co/t/project04371/m/project04371/1770704602624/10404d0ba90a4298a0d03348bbbc3db976651c79

Brainstorm and idea prioritization

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Category Analysis

Time-based Analysis

Visualization and Dashboard

Special Analysis

Heritage Site Distribution

Category Analysis

Time-based Analysis

Visualization and Dashboard

Special Analysis

Importance

If each of these ideas could get done without any difficulty or cost, which would have the most positive impact?

12%

After you collaborate
You can export the mural as an image or pdf to share with members of your company who might find it helpful.

Quick add-ons

- Show the mural**
Share a view link to the mural with collaborators to keep them in the loop about the outcomes of the session.
- Export the mural**
Export a copy of the mural as a PNG or PDF to attach to emails, include in decks, or save in your drive.

Keep moving forward

- Strategy blueprint**
Define the components of a new idea or strategy.
[Open the template](#)
- Customer experience journey map**
Understand customer needs, evaluations, and obstacles for an experience.
[Open the template](#)
- Strengths, weaknesses, opportunities & threats**
Identify strengths, weaknesses, opportunities, and threats (SWOT) to develop a plan.
[Open the template](#)

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

Scenario: UNESCO World Heritage Professional Uses the dashboard to monitor heritage sites, assess risk status, and support conservation decisions.	 Entice How does someone become aware of this service?	 Enter What do people experience as they begin the process?	 Engage In the core moments in the process, what happens?	 Exit What do people typically experience as the process finishes?	 Extend What happens after the experience is over?
 Experience steps What does the person (or people) at the center of this scenario typically experience in each step?	Hears about the dashboard through reports or colleagues	Opens dashboard and explores filters	Analyzes country, region, category, and danger status	Uses insights to prepare reports or decisions	Revisits dashboard for updates
 Interactions What interactions do they have at each step along the way? - People: Who do they see or talk to? - Places: Where are they? - Things: What digital touchpoints or physical objects do they use?	Colleagues, UNESCO portal, emails	Dashboard UI, filters, maps	Charts, maps, hover tooltips	Exports, screenshots, meetings	Updated data, shared dashboards
 Goals & motivations At each step, what is a person's primary goal or motivation? ("Help me..." or "Help me avoid...")	Find reliable heritage data quickly	Understand site distribution and risks	Identify high-risk sites and priority regions	Support conservation planning	Track progress over time
 Positive moments What steps does a typical person find enjoyable, productive, fun, motivating, delightful, or exciting?	Centralized global data	Clean visuals, easy navigation	Clear comparisons, interactive insights	Confident, data-backed decisions	Consistent monitoring
 Negative moments What steps does a typical person find frustrating, confusing, angering, costly, or time-consuming?	Unsure about data accuracy	Too many filters initially	Overlapping map points	Limited annotations	Manual refresh needed
 Areas of opportunity How might we make each step better? What ideas do we have? What have others suggested? Product School: Online opportunity with the product	Clear intro and data source explanation	Guided walkthrough or tooltips	Highlight critical sites and trends	Add summary insights and notes	Automated updates and alerts

3.2 Solution Requirement

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Visualization	Display UNESCO World Heritage Sites data using charts and maps
FR-2	Country Analysis	Show number of heritage sites by country
FR-3	Regional Analysis	Display site distribution across regions
FR-4	Category Analysis	Visualize Cultural, Natural, and Mixed site categories
FR-5	Trend Analysis	Show year-wise inscription trends using line charts
FR-6	Risk Analysis	Display sites at risk (In Danger vs Not in Danger)
FR-7	Interactive Filters	Filter data by Year, Region, Country, and Category
FR-8	Map Visualization	Show geographical distribution of sites on maps

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard should be simple, intuitive, and easy to navigate for all users.
NFR-2	Security	The dashboard should ensure data integrity and prevent unauthorized modification of data.
NFR-3	Reliability	The dashboard should consistently display accurate and correct information without errors.
NFR-4	Performance	The dashboard should load quickly and respond smoothly to filters and interactions.
NFR-5	Availability	The dashboard should be accessible whenever required without downtime.
NFR-6	Scalability	The dashboard should handle increasing data size without performance degradation.

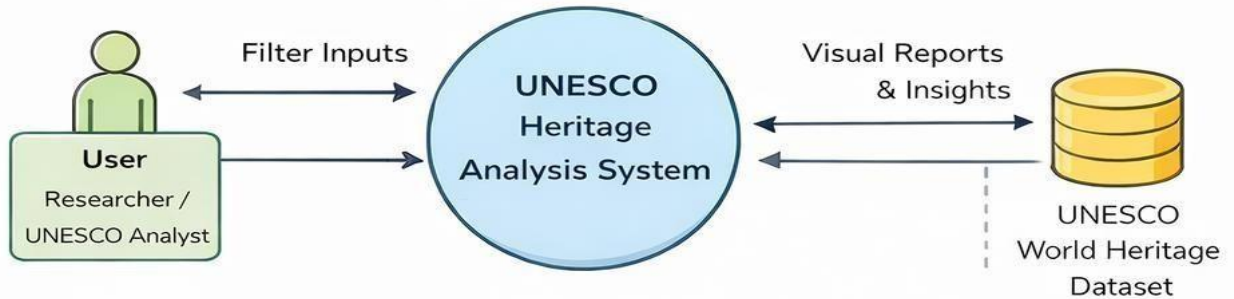
3.3 Data Flow Diagram Data

Flow Diagrams:

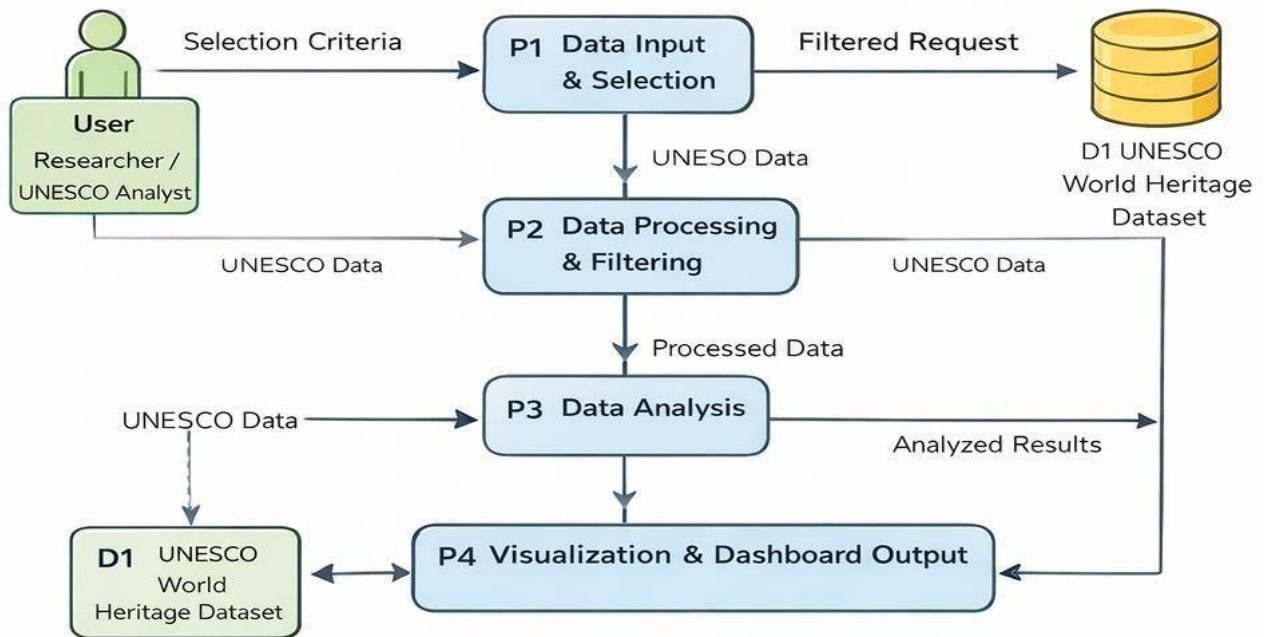
Data Flow Diagram (DFD)

Heritage Treasures – UNESCO World Heritage Sites Analysis System

DFD Level 0 – Context Diagram



DFD Level 1 – Detailed Diagram



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
UNESCO Professional	Login	USN-1	As a UNESCO professional, I can log in to access the dashboard.	User can log in successfully	High	Sprint-1
UNESCO Professional	Dashboard Overview	USN-2	As a user, I can view total heritage sites by country and region.	Country & region counts are visible	High	Sprint-1
UNESCO Professional	Risk Monitoring	USN-3	As a user, I can see sites marked In Danger and Not in Danger.	Risk status charts display correctly	High	Sprint-1
UNESCO Professional	Category Analysis	USN-4	As a user, I can analyse sites by category (Cultural, Natural, Mixed).	Category-wise charts load	Medium	Sprint-1
UNESCO Professional	Trend Analysis	USN-5	As a user, I can view yearwise inscription trends.	Line chart shows yearly trends	Medium	Sprint-1
UNESCO Professional	Geographic Visualization	USN-6	As a user, I can view heritage sites on an interactive world map.	Map shows accurate locations	High	Sprint-2
UNESCO Professional	Area Comparison	USN-7	As a user, I can compare heritage site areas (hectares) by country.	Filled map displays area size	Low	Sprint-2
UNESCO Professional	Advanced Filters	USN-8	As a user, I can filter data by year, category, region, and country.	Filters update visuals dynamically	Medium	Sprint-2
UNESCO Professional	Insight Exploration	USN-9	As a user, I can explore less prominent countries with fewer sites.	Data is visible for all countries	Low	Sprint-3

3.4 Technology Stack

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Technical Architecture

UNESCO World Heritage Sites Dashboard

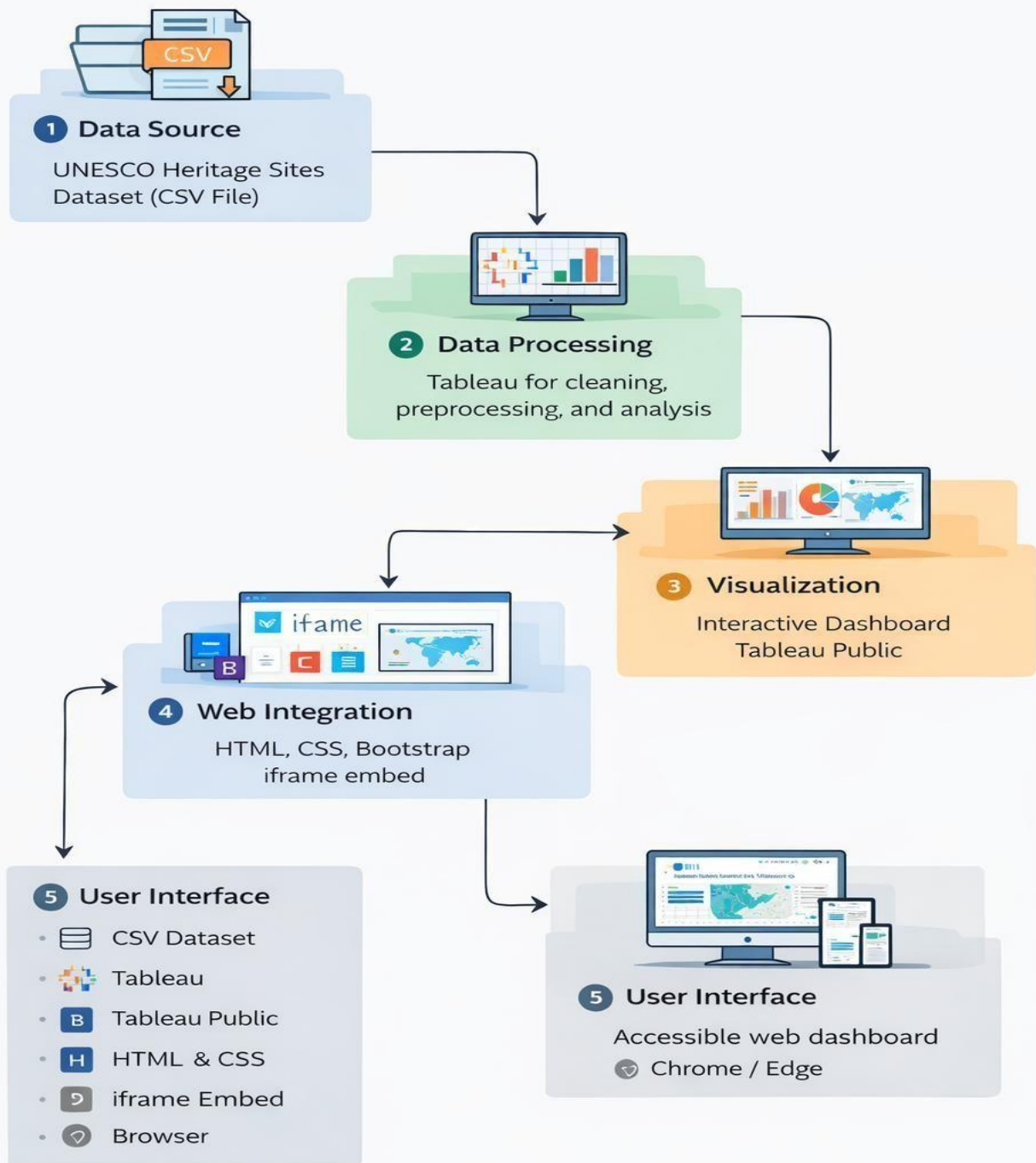


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web interface through which users view dashboards, stories, maps, and insights	HTML, CSS, JavaScript, Bootstrap

2.	Frontend Logic	Handles page navigation, UI interactions, responsiveness, and animations	JavaScript, Bootstrap JS
3.	Web Framework	Connects frontend with backend and manages routing and page rendering	Flask (Python)
4.	Application Logic	Controls application flow and serves embedded dashboards and static content	Python (Flask)
5.	Data Visualization Engine	Creates interactive charts, maps, dashboards, and stories	Tableau Public
6.	Data Processing Layer	Performs data cleaning, aggregation, filtering, and calculations	Tableau Calculated Fields
7.	Dataset	Stores UNESCO World Heritage data such as country, region, category, year, and danger status	CSV / Excel Dataset
8.	Embedding Layer	Embeds Tableau dashboards and stories into web pages	Tableau Embed
9.	Static Resource Management	Stores CSS, JS, images, icons, and template assets	Flask Static Folder
10.	Infrastructure (Server / Cloud)	Hosts and runs the Flask web application	Local System / Cloud Hosting

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
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1.	Open-Source Frameworks	Uses open-source frameworks and libraries for building the web interface and backend server	Flask (Python), Bootstrap, JavaScript
2.	Security Implementations	No user authentication is required. The application follows basic web security practices such as secure embedding and safe routing	HTTPS, Secure Embedding
3.	Scalable Architecture	Follows a layered architecture where frontend, backend, and visualization layers are loosely coupled, allowing easy scalability	Flask Backend, Tableau Public
4.	Availability	Application is accessible anytime as dashboards are hosted on Tableau Public and served through a Flask web server	Tableau Public Hosting, Flask Server
5.	Performance	Optimized by using pre-aggregated data in Tableau dashboards and lightweight frontend rendering	Tableau Optimized Views, Bootstrap

4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem-Solution fit canvas 2.0			Purpose / Vision	
Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS Students studying history, archaeology, geography, and data analytics Researchers and academicians analyzing UNESCO World Heritage data General users interested in global heritage sites and insights	6. CUSTOMER CONSTRAINTS CC Difficulty in understanding raw or unstructured data Lack of interactive tools in traditional reports Time constraints in manual analysis	5. AVAILABLE SOLUTIONS AS Static reports and PDFs from UNESCO website Raw datasets without visualization Individual charts without integrated dashboards	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&P Understand the global distribution of UNESCO World Heritage Sites Analyze sites by country, region, category, and danger status Identify historical trends and patterns in heritage site inscriptions	9. PROBLEM ROOT CAUSE RC UNESCO data is available but not presented in an interactive and user-friendly way Lack of centralized visual dashboards for easy analysis	7. BEHAVIOUR BE Users search online for heritage-related statistics Compare countries and regions manually using spreadsheets Rely on static images or textual summaries	
Identify strong TR & EM	3. TRIGGERS TR Academic projects and assignments related to data visualization Curiosity about cultural and natural heritage around the world Need for quick insights instead of reading large reports	10. YOUR SOLUTION SL An interactive UNESCO World Heritage Dashboard built using Flask, Bootstrap, and Tableau Provides clear visual insights such as country-wise, region-wise, category-wise analysis Enables easy understanding of trends, distribution, and danger status without technical expertise	8. CHANNELS of BEHAVIOUR CH 8.1 Online Educational websites and dashboards Tableau Public visualizations Web browsers on desktop and mobile	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER EM Before: Confused, overwhelmed by raw data and large datasets After: Confident, informed, and satisfied through clear visual insights		8.2 Offline Classroom discussions Academic presentations and reports	

4.2 Proposed Solution

Proposed Solution Template:

Project team shall fill the following information in proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	UNESCO World Heritage data is available in raw and report formats, making it difficult for students and researchers to quickly analyse trends, regional distribution, categories, and danger status of heritage sites.
2.	Idea / Solution description	Develop an interactive web-based dashboard using Flask, Bootstrap, and Tableau to visualize UNESCO World Heritage Sites data. The dashboard provides country-wise, region-wise, category-wise, year-wise, and danger status analysis in a simple and interactive format.
3.	Novelty / Uniqueness	- Combines data visualization + storytelling in one platform. - Converts complex heritage datasets into easy-to-understand visuals. - User-friendly academic dashboard without login requirement.

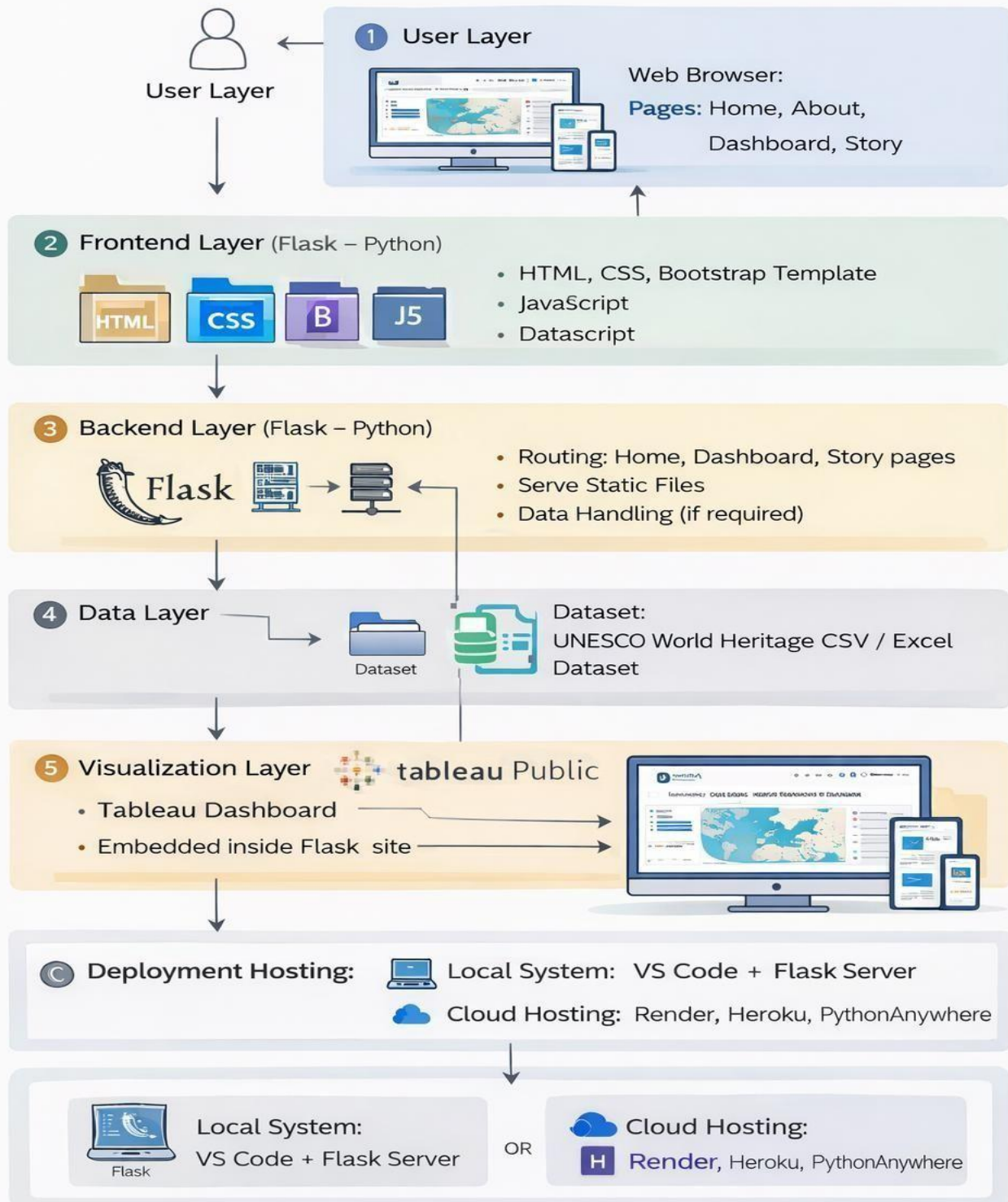
4.	Social Impact / Customer Satisfaction	- Helps students and researchers understand global cultural and natural heritage. - Encourages awareness about endangered heritage sites. - Supports academic learning and research projects.
5.	Business Model (Revenue Model)	- Academic and educational use (non-commercial). - Can be extended as a premium research analytics tool. - Potential integration with tourism analytics platforms.
6.	Scalability of the Solution	- Can be extended to include real-time UNESCO updates. - Can integrate more datasets (tourism data, environmental data). - Deployable on cloud platforms for large-scale usage.

4.3 Solution Architecture

Solution Architecture Diagram:

Solution Architecture

UNESCO World Heritage Dashboard



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Product Backlog, Sprint Schedule, and Estimation:

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Dataset Preparation	USN-1	As a user, I want cleaned UNESCO dataset so that I can analyse accurate information.	5	High	Team Leader
Sprint-1	Backend Setup (Flask)	USN-2	As a developer, I want to integrate Flask backend to serve dashboard pages.	5	High	Team Leader
Sprint-1	Dashboard Design	USN-3	As a user, I want an interactive dashboard showing country-wise site counts.	10	High	Team Leader
Sprint-2	Regional Analysis	USN-4	As a user, I want region-wise comparison of heritage sites.	8	High	Team Leader
Sprint-2	Category Analysis	USN-5	As a user, I want category-wise distribution (Cultural, Natural, Mixed).	4	Medium	Team Leader
Sprint-2	Danger Status Analysis	USN-6	As a user, I want to see sites in danger vs not in danger.	8	High	Team Leader
Sprint-3	Map Visualization	USN-7	As a user, I want a world map showing heritage site distribution.	6	High	Team Leader
Sprint-3	Year-wise Trend Analysis	USN-8	As a user, I want inscription trends over years.	4	Medium	Team Leader
Sprint-3	Web Integration	USN-9	As a user, I want the dashboard embedded in a responsive website.	10	High	Team Leader

Project Tracker, Velocity & Burndown Chart:

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	5 Days	03 Feb 2026	07 Feb 2026	20	07 Feb 2026
Sprint-2	20	5 Days	08 Feb 2026	12 Feb 2026	20	12 Feb 2026
Sprint-3	20	5 Days	13 Feb 2026	17 Feb 2026	20	17 Feb 2026

Velocity:

Velocity per Sprint: = 20 Story Points

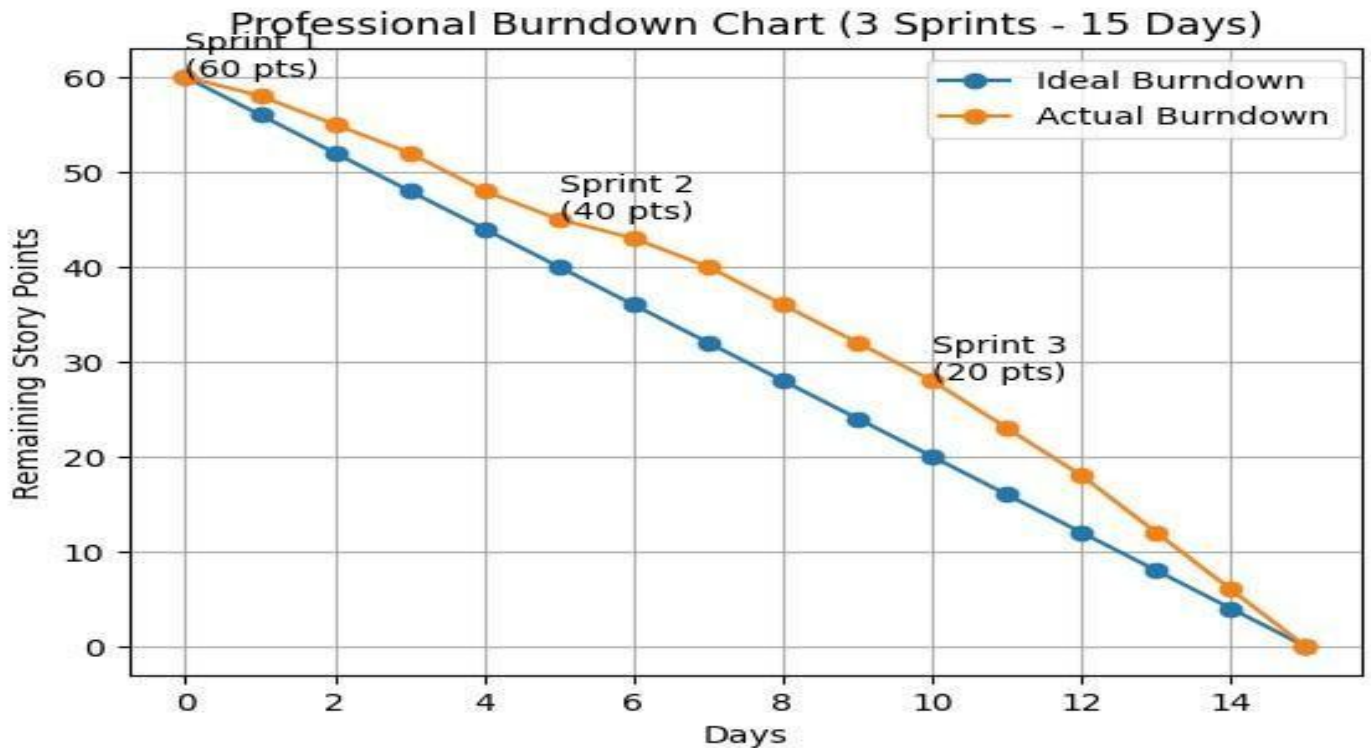
Average Velocity: $(20+20+20) \div 3 = 20$

Velocity per Day: Team Velocity = 20 points per sprint
Each sprint = 5 days
 $20 \div 5 = 4$

Average Velocity = 4 story points per day Burndown

Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.



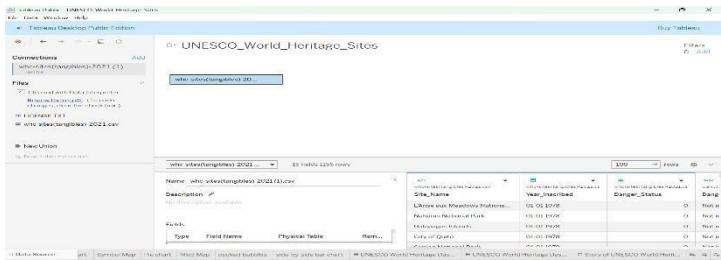
6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Screenshot / Values
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1.	Data Rendered	 whc-sites(tangibles)-2021 ... 15 fields 1155 rows Data visualized using 6 interactive charts and 2 maps in the dashboard.
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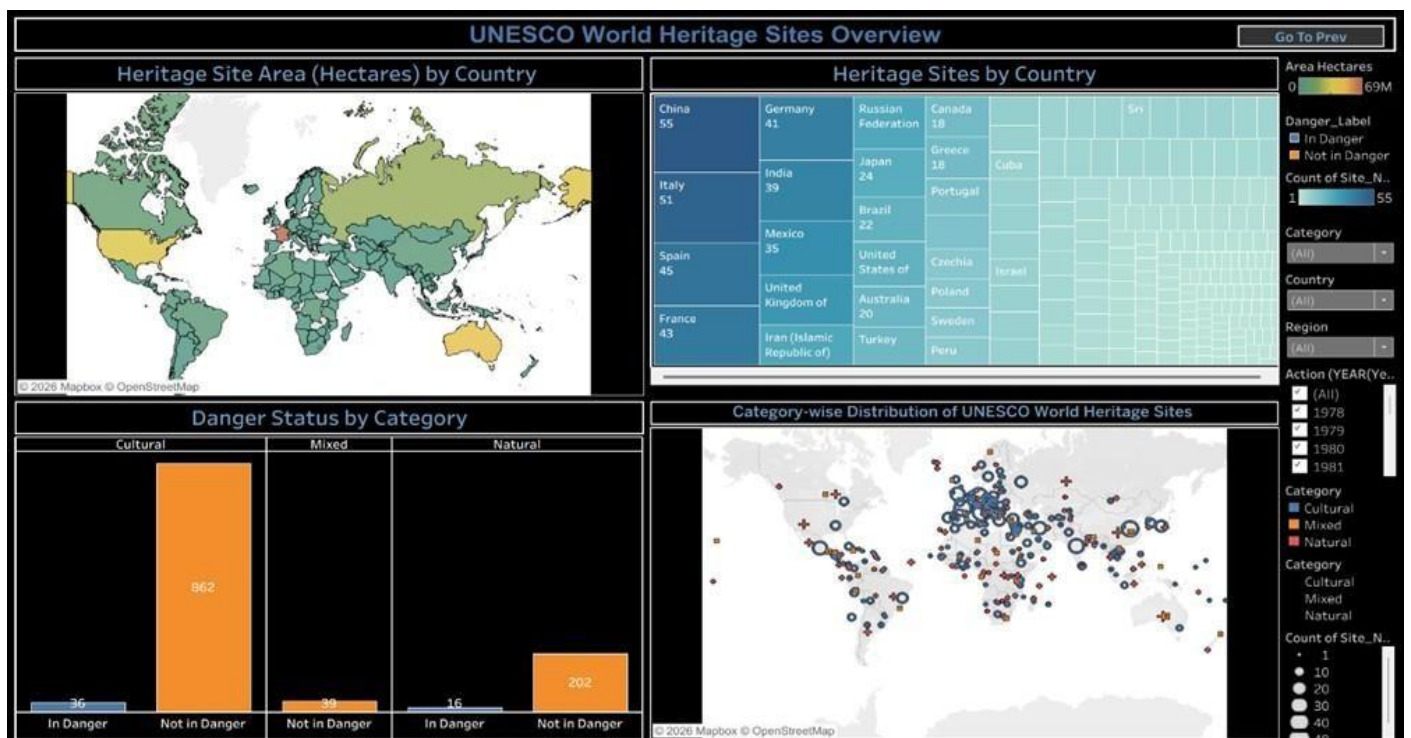
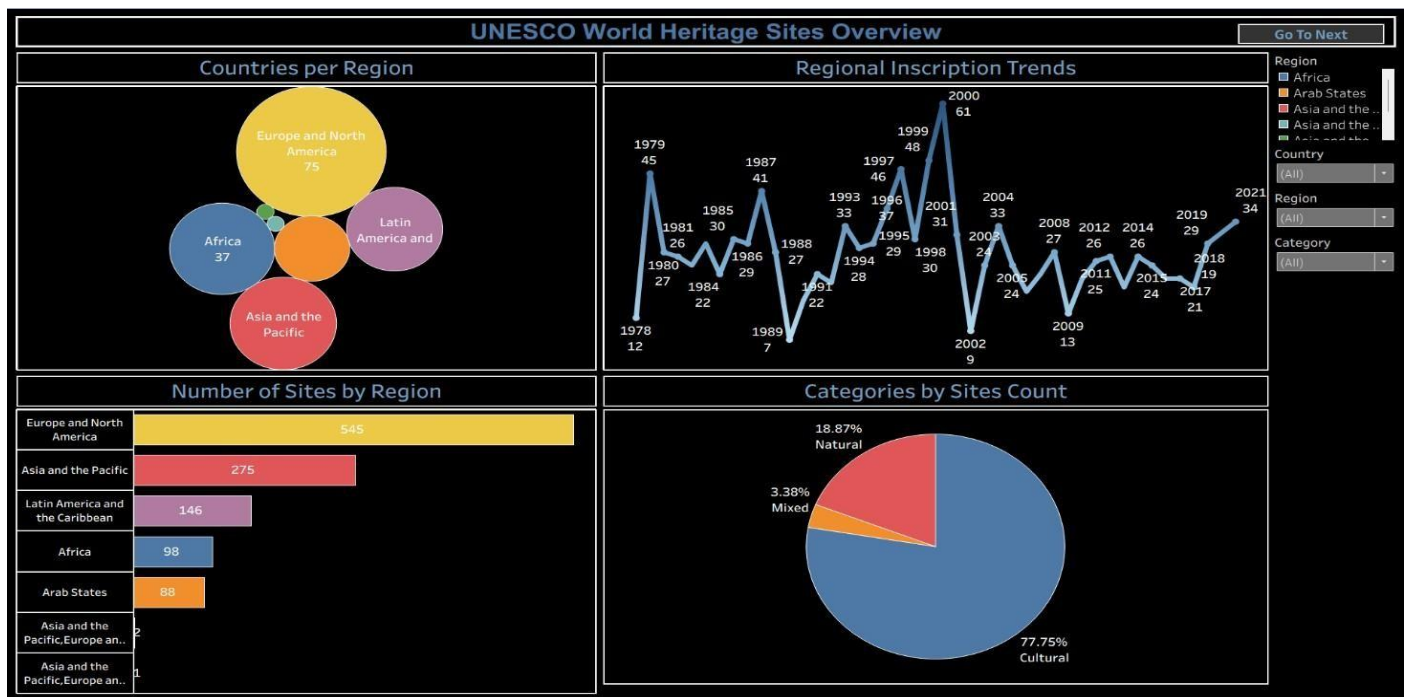
2.	Data Preprocessing	- Renamed dataset to “UNESCO World Heritage Sites” - Renamed columns for clarity: - name → Site_Name - country name → Country - date_inscribed → Year_Inscribed - danger → Danger_Status - category_long → Category - Cleaned column headers using Data Interpreter - Fixed data types: Site_Name → String (Text) Country → String (Text) Region → String (Text) Category → String (Text) Year_Inscribed → Integer (Number) Danger_Status → NUmber(Whole) Area_Hectares → Number (Decimal)
3.	Utilization of Filters	Created Filters: - Region Filter - Country Filter - Category Filter - Year (Action Filter) - Interactive navigation (Go to Next / Previous dashboard)
4.	Calculation fields Used	Created Filters: - Danger_Label - Site_Status - Transboundary_Label - Data_End_Status

5.	Dashboard design	No of Visualizations / Graphs – 8 Dashboard_Page1: • Countries per Region (Bubble Chart) • Regional Inscription Trends (Line Chart) • Number of Sites by Region (Horizontal Bar Chart) • Categories by Sites Count (Pie Chart) Dashboard_Page2: • Heritage Site Area(Hectares) by Country (Map) • Heritage Sites by Country (Tree map) • Danger Status by Category (Side-By-Side Bar Chart) • Category-Wise Distribution of UNESCO World Heritage Sites (Symbol Map)
6	Story Design	No of Visualizations / Graphs -8 Stories Created: • Countries per Region (Bubble Chart) • Regional Inscription Trends (Line Chart) • Number of Sites by Region (Horizontal Bar Chart) • Categories by Sites Count (Pie Chart) • Heritage Site Area(Hectares) by Country (Map) • Heritage Sites by Country (Tree map)
		• Danger Status by Category (Side-By-Side Bar Chart) • Category-Wise Distribution of UNESCO World Heritage Sites (Symbol Map)

7. RESULTS 7.1

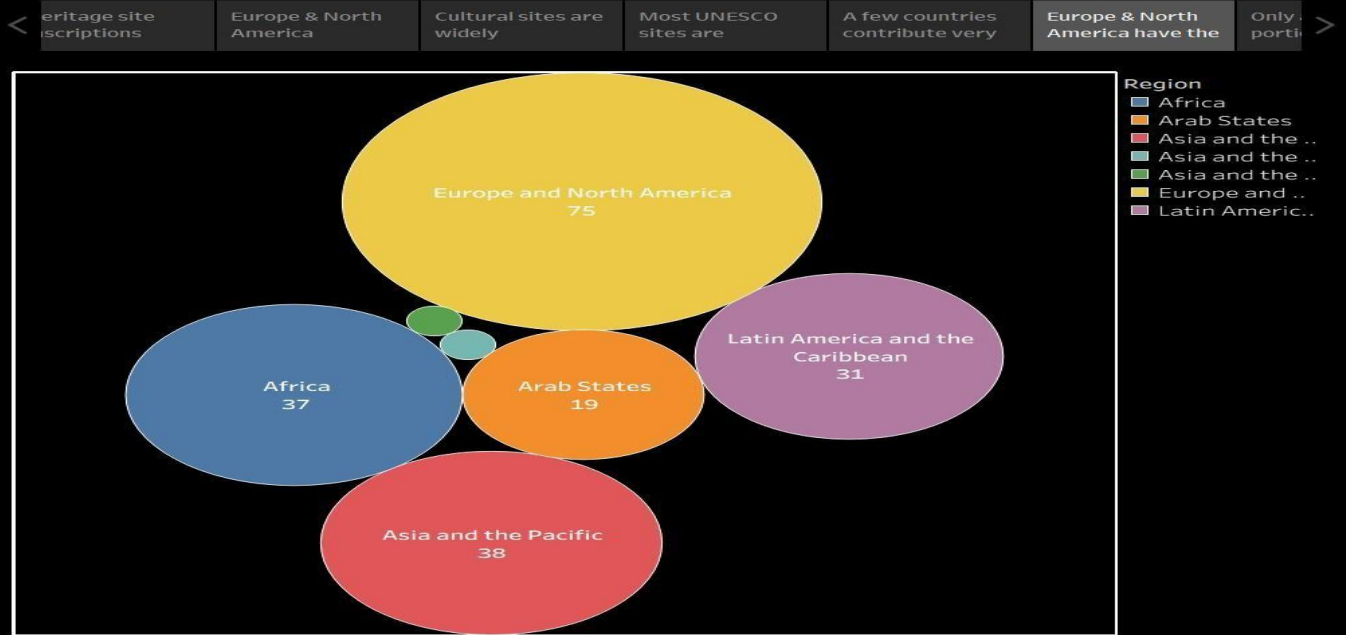
Output Screenshots

Dashboard Screenshots:

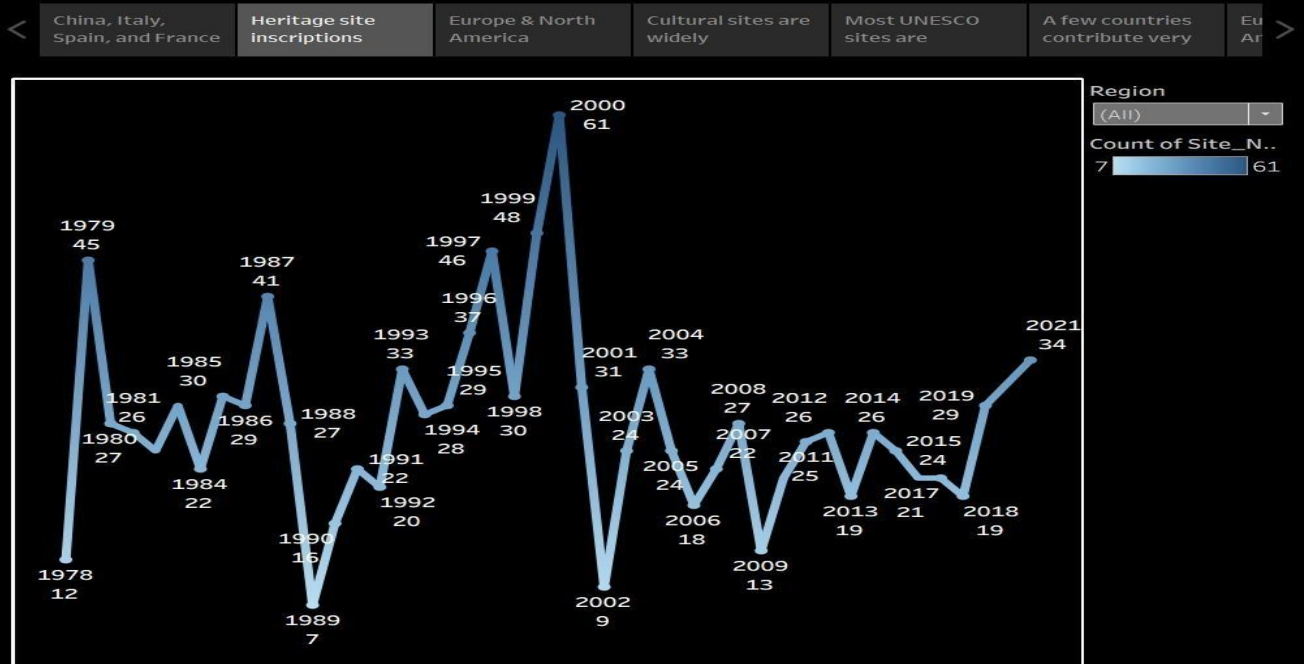


Story Screenshots:

Story of UNESCO World Heritage Sites Visualizations

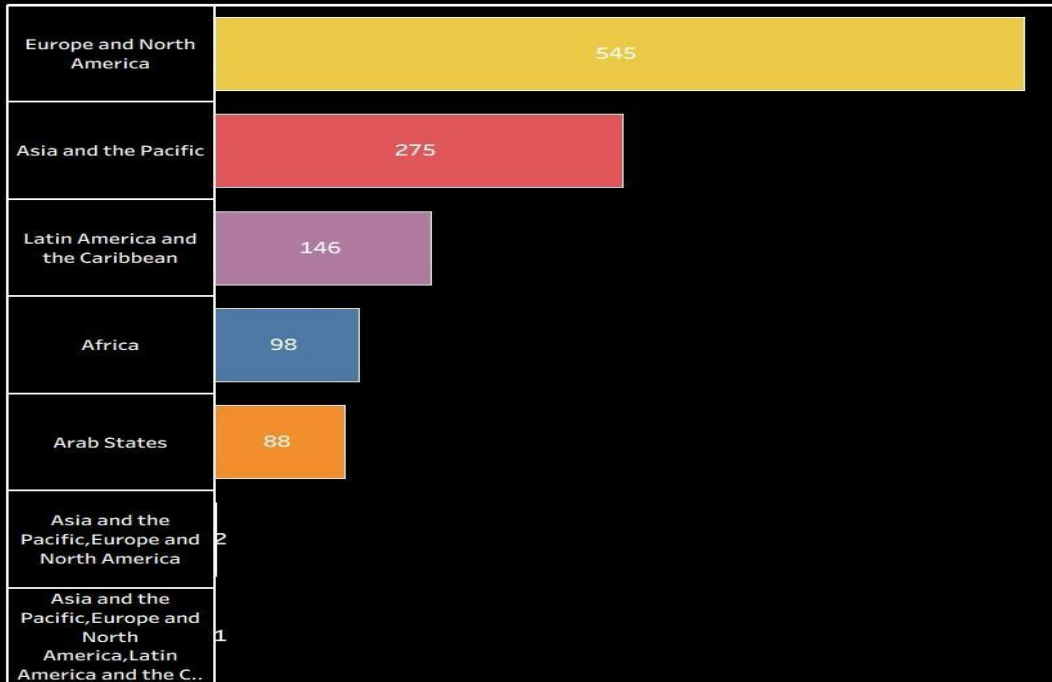


Story of UNESCO World Heritage Sites Visualizations



Story of UNESCO World Heritage Sites Visualizations

[China, Italy, Spain, and France](#)
[Heritage site inscriptions](#)
[Europe & North America](#)
[Cultural sites are widely](#)
[Most UNESCO sites are](#)
[A few countries contribute very](#)
[Europe and North America](#)



Region

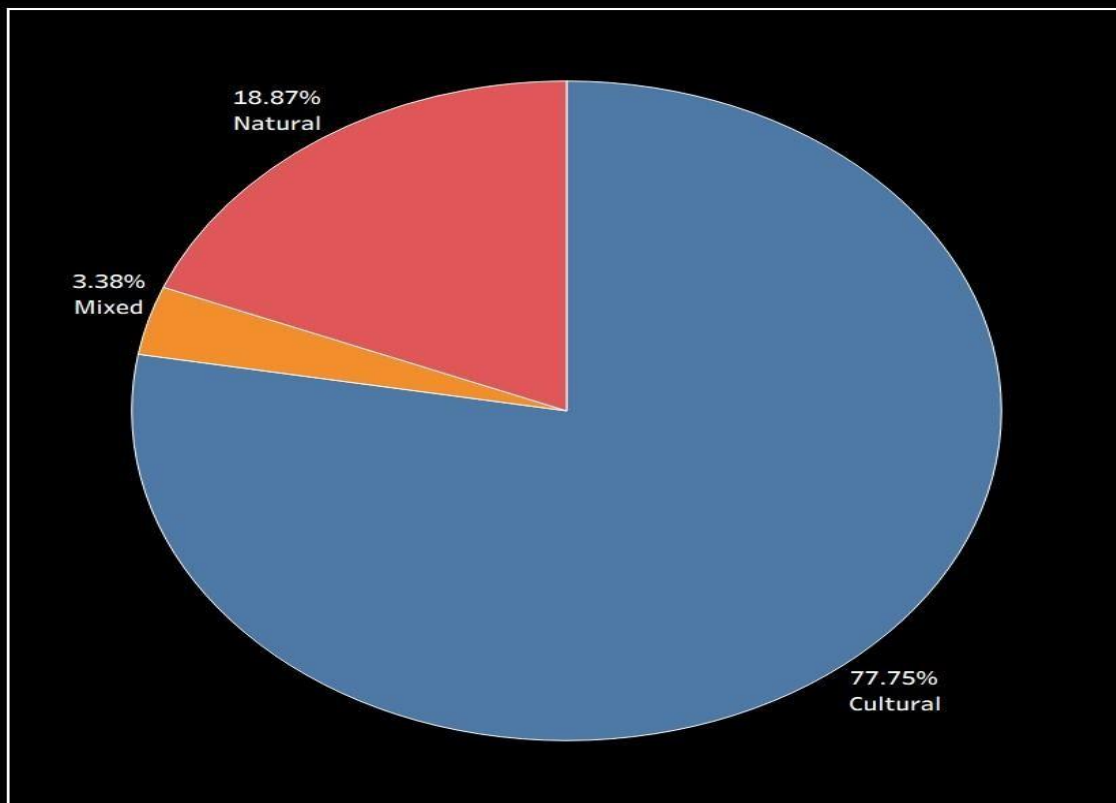
(All)

Region

- Africa
- Arab States
- Asia and the Pacific
- Asia and the Pacific, Europe and North America
- Asia and the Pacific, Europe and North America, Latin America and the Caribbean
- Europe and North America
- Latin America and the Caribbean

Story of UNESCO World Heritage Sites Visualizations

[China, Italy, Spain, and France](#)
[Heritage site inscriptions](#)
[Europe & North America](#)
[Cultural sites are widely](#)
[Most UNESCO sites are](#)
[A few countries contribute very](#)
[Europe and North America](#)



Action (YEAR(Ye..

(All)

Region

(All)

Category

- Cultural
- Mixed
- Natural

Story of UNESCO World Heritage Sites Visualizations

China, Italy, Spain, and France

Heritage site inscriptions

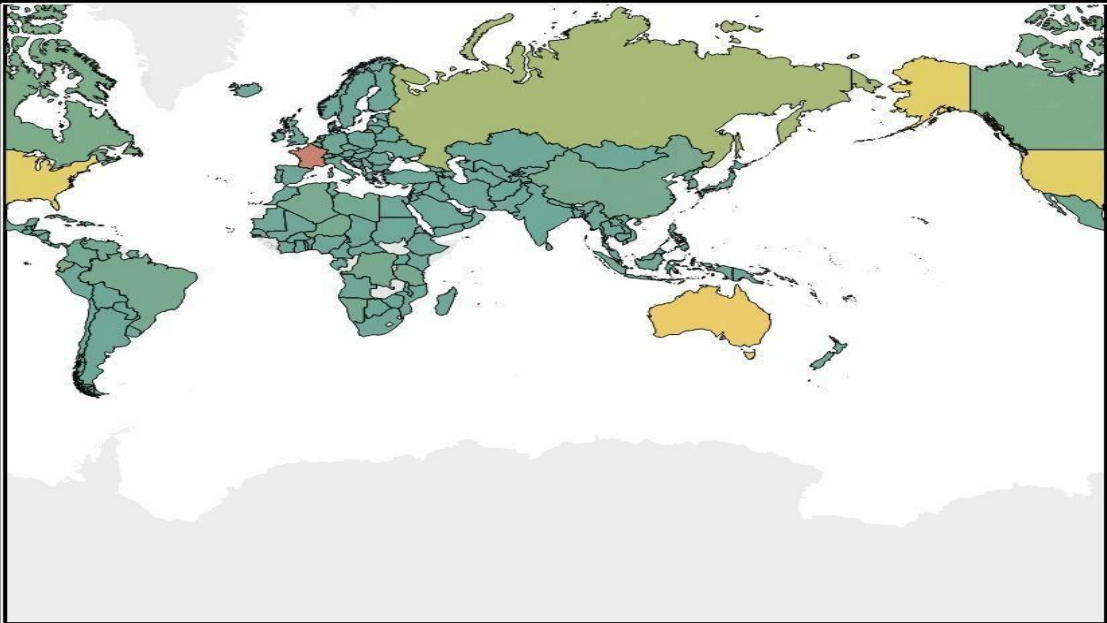
Europe & North America

Cultural sites are widely

Most UNESCO sites are

A few countries contribute very

Eu
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© 2026 Mapbox © OpenStreetMap

Story of UNESCO World Heritage Sites Visualizations

China, Italy, Spain, and France

Heritage site inscriptions

Europe & North America

Cultural sites are widely

Most UNESCO sites are

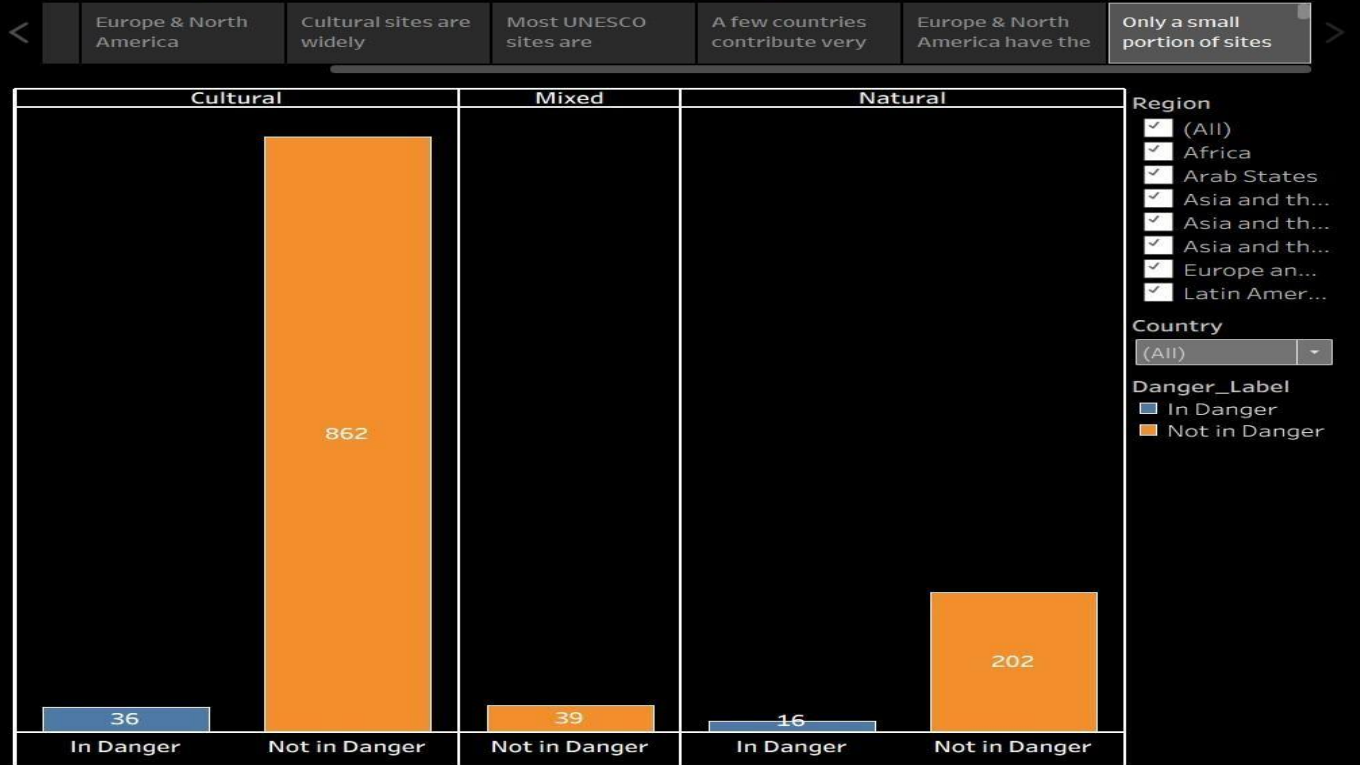
A few countries contribute very

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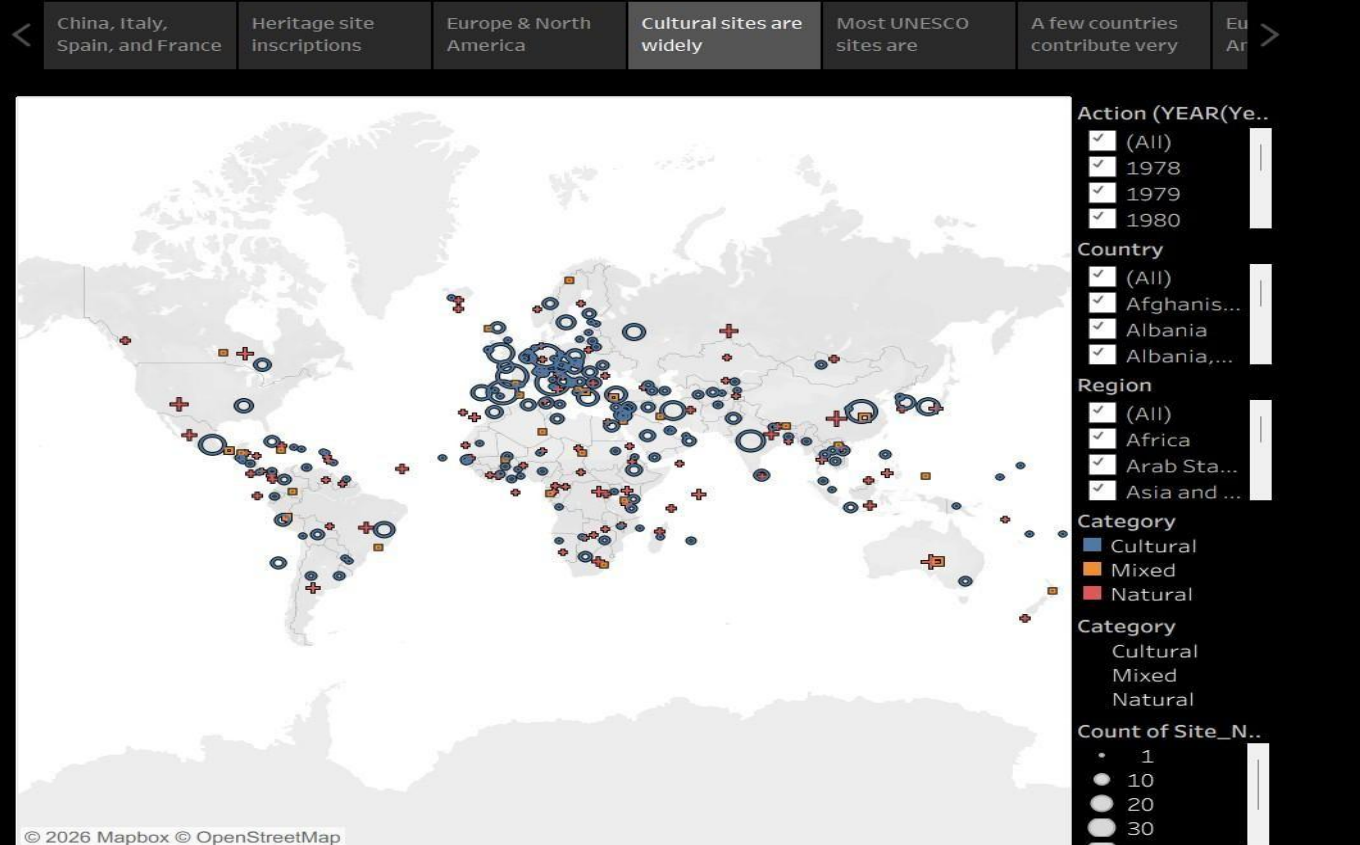
China 55	Mexico 35	United Kingdom of Great Britain and Northern Ireland	Iran (Islamic Republic of) 26	Russian	Japan 24	Brazil 22	United States of
Italy 51	Australia 20		Peru 12				
Spain 45	Turkey 19					Sri	Viet
France 43	Canada 18	Algeria	Iraq				
Germany 41	Greece 18	Austria	Jordan				
	Portugal 16	Croatia					
	Republic of Korea	Egypt					
	Czechia 13	Kenya					
	Poland 13	Norway					
		Chile					

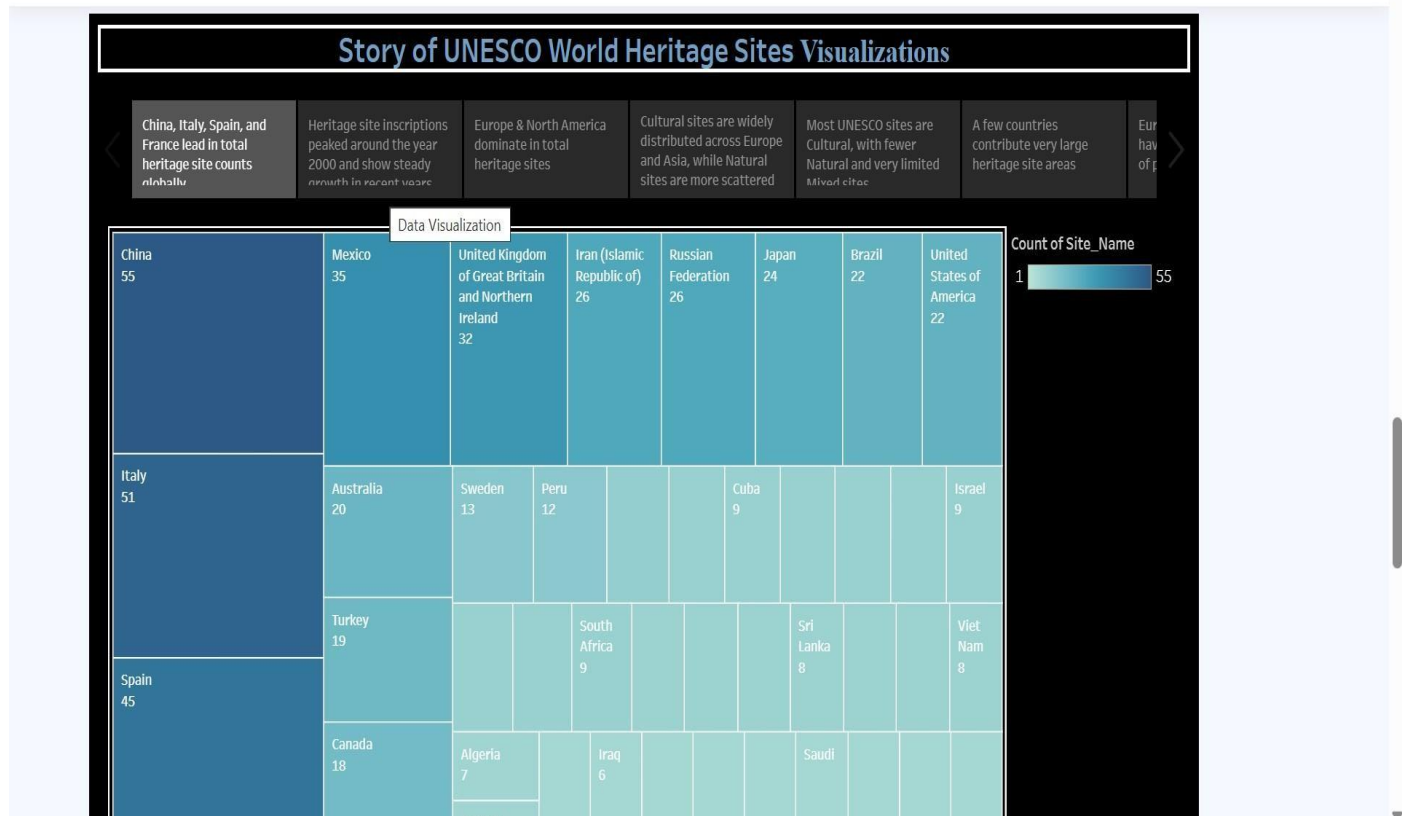
Count of Site_N..
1 55

Story of UNESCO World Heritage Sites Visualizations



Story of UNESCO World Heritage Sites Visualizations





8. ADVANTAGES & DISADVANTAGES **Advantages:**

- Provides clear visual understanding of UNESCO World Heritage data.
- Interactive dashboard with filters (Region, Country, Category, Year).
- Easy comparison of regions and countries.
- Helps identify trends and patterns over time.
- Integrated with web application using Flask for accessibility.
- Useful for academic research and data analysis.

Disadvantages:

- Depends on the accuracy of the dataset.
- Static dataset (2021) – not real-time updated.
- Limited advanced predictive analytics.
- Performance may reduce with very large datasets.

9. CONCLUSION

The UNESCO World Heritage Sites Dashboard successfully visualizes global heritage data in an interactive and meaningful way. The project demonstrates how data preprocessing, visualization, and web integration can transform raw data into actionable insights. It helps users understand site

distribution, regional trends, and danger status effectively. Overall, the project achieves its objective of providing a clear analytical view of UNESCO heritage data.

10. FUTURE SCOPE

- Integrate real-time updated UNESCO dataset.
- Add predictive analysis using Machine Learning.
- Include tourism statistics and visitor data.
- Add advanced analytics such as growth forecasting.
- Improve UI/UX design and performance optimization.
- Deploy the project on cloud for global accessibility.
- Add user authentication and report download features.